MERN STACK

DAY – 32

31 July 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

ADMIN DASHBOARD

• The Admin Dashboard is developed to provide administrators with complete control and monitoring of the ParkEasy parking management system.
• It helps administrators view parking statistics, user information, and booking activities.
• The useState() hook is used to store dashboard statistics received from the backend.
• The useEffect() hook is used to fetch admin dashboard data automatically when the page loads.
• The Sidebar component is integrated to provide navigation options for the admin panel.

• The Admin Dashboard fetches data from the backend using the API:

GET /admin-dashboard

• The dashboard displays the following information:

1. Total Parking Slots
2. Available Slots
3. Reserved Slots
4. Maintenance Slots
5. Total Users
6. Pending Booking Requests

• The Parking Overview section displays:

1. Approved Bookings

2. Cancelled Bookings
3. Current Vehicles Parked

• The dashboard data is dynamically displayed using React state values.
• The interface helps administrators monitor the complete parking system efficiently.

ADMIN DASHBOARD API DEVELOPMENT

• The Admin Dashboard API is created using Express.js to provide overall system statistics.
• The API retrieves user and parking slot information from MongoDB.

API Endpoint:

```
GET /admin-dashboard
```

• All user records are fetched using:

```
User.find()
```

• All parking slot records are fetched using:

```
Slot.find()
```

• The API calculates parking details by checking booking records from users.

• Booking statuses are calculated:

- Pending Bookings
- Approved Bookings
- Cancelled Bookings
- Current Parking Vehicles

• Slot information is calculated:

- Total Slots
- Available Slots

- Reserved Slots
- Maintenance Slots

Working

1. The AdminDashboard() component is created to display admin statistics.
2. The component sends a request to the Admin Dashboard API using Fetch API.
3. The backend retrieves all users and parking slot details from MongoDB.
4. Booking records are checked to calculate different statuses.
5. The API sends complete dashboard information to the frontend.
6. The received data is stored using React state.
7. The administrator can monitor parking activities through dashboard cards.

```
app.get("/admin-dashboard", async(req,res)=>{

try{
const users = await User.find();
const slots = await Slot.find();
let pendingBookings = 0;
let approvedBookings = 0;
let cancelledBookings = 0;
let currentParking = 0;
users.forEach(user=>{
user.bookings.forEach(booking=>{
if(booking.status==="Pending")
pendingBookings++;
if(booking.status==="Approved")
approvedBookings++;
if(booking.status==="Cancelled")
cancelledBookings++;
if(booking.status==="Approved")
currentParking++;
});
});
res.json({
totalSlots:slots.length,
availableSlots:
slots.filter(s=>s.status==="Available").length,
reservedSlots:
slots.filter(s=>s.status==="Reserved").length,
maintenanceSlots:
slots.filter(s=>s.status==="Maintenance").length,
totalUsers:users.length,

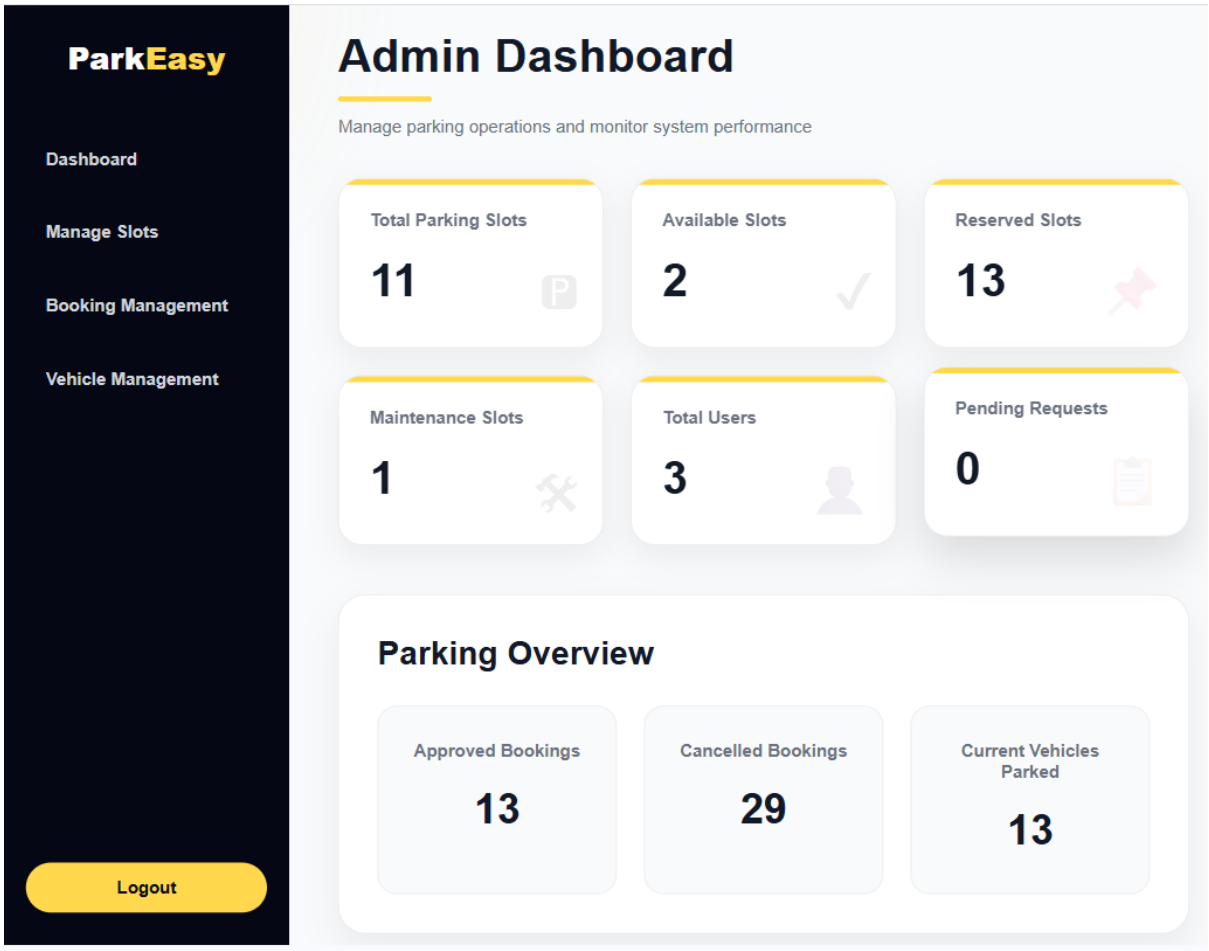
pendingBookings,

approvedBookings,

cancelledBookings,

currentParking

});
}
catch(err){
res.status(500).json({
message:err.message
});
}
});
```



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